



DeepLeuk: a convolutional neural network pre-trained model for microscopic cell images-Based leukemia Cancer analysis

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Abstract

The blood and bone marrow are affected by leukemia, a deadly kind of cancer, that significantly impacts the quality of life of those diagnosed. Early identification and precise diagnosis are crucial for improving survival rates. Fortunately, recent advancements in medical image analysis, particularly deep learning-based techniques, have greatly improved the ability to distinguish leukemia cells from healthy ones through microscopic cell images. This research introduces a deep learning-based leukemia cancer classifier, specifically a CNN pre-trained model, utilizing microscopic cell images to detect malignant cells. Using pre-processing techniques such as picture scaling, Region of Interest (RoI) extraction, and Improved Anisotropic Filtering (IAF) and feature extraction, the blood cell image dataset is first cleaned. After that leukemia-affected and healthy cells are evaluated using various classification algorithms and neural networks, with optimal features identified to improve classifier performance. The results suggest that neural networks function well as a classifier algorithm to detect whether the person is cancerous or non-cancerous, with the proposed CNN pre-trained model providing precision of 98.9%, which is higher than any other method mentioned. The proposed model prioritizes recall, a key performance metric, to reduce the number of false negatives. Accurate diagnosis and treatment are critical, as misdiagnosing a patient with cancer as not having cancer can lead to severe consequences. With the main objective of minimizing inadvertent mistakes made by physicians, the proposed model performs better than kNN, Decision Trees, Random Forest, SVM, and Logistic Regression models. Using deep learning-based techniques to improve cancer diagnosis and treatment is essential. Improving survival rates and the quality of life for individuals with leukemia requires early identification and accurate diagnosis. This research can help doctors make more accurate diagnoses, leading to more effective treatments and better outcomes for patients.

Keywords Leukemia Cancer classifier · CNN pre-trained model · Convolutional neural networks · Microscopic cell images · Image classification · Diagnosis

Extended author information available on the last page of the article

1 Introduction

GLOBOCAN (<http://globocan.iarc.fr>) predicts that 25,000 children in India are diagnosed with cancer annually, around 9000 of them with leukemia. In ten years, 90,000 Indian youngsters would get leukemia, even using these conservative predictions. An estimated 440,000 new instances of leukemia are identified worldwide each year, and 310,000 people die from the disease. In 2023, there will be over 24,000 leukemia-related fatalities and an estimated 60,000 new instances of the disease in the US. Leukemia still poses a serious risk to human health, although the outlook is not good right now. In 2020, leukemia was the 11th most prevalent cause of cancer-related mortality worldwide and the 15th most common cause of cancer incidence, with 474,519 incident cases and 311,594 fatalities. In 2020, there were 474,519 newly reported cases of leukemia. With a variation of over five times, the age-standardized incidence rate was 5.4 per 100,000 worldwide.

The unchecked proliferation and division of aberrant cells in the body characterize cancer, a chronic illness. Cancer can show various symptoms depending on where it is and how advanced it is. White blood cell production that is abnormally high is a sign of leukaemia, a blood and bone marrow malignancy. This condition occurs when the DNA of immature blood cells, known as blasts, becomes damaged and grows uncontrollably, leading to fatigue, infections, and bleeding. While the exact causes of leukemia are unknown, risk factors such as radiation exposure, chemical exposure, and hereditary factors may contribute to its development. Early detection of leukemia is crucial for better treatment options, as early-stage cancers are generally easier to treat and may require less aggressive interventions compared to advanced-stage cancers. Timely detection increases the chances of successful treatment and improves the prospects of a cure.

Deep learning for leukaemia identification has been the subject of recent study, with encouraging outcomes. AML, CLL, CML, and ALL are among the leukaemia subtypes that deep learning algorithms can automatically recognize and categorize. These models can utilize large datasets of medical images, extract relevant features, and accurately classify the type and severity of leukemia. Additionally, deep learning algorithms can continuously improve performance with more data, making them highly adaptable and scalable for leukemia detection. Deep learning has the potential to minimise human variability and mistakes, which is one of the main benefits of utilizing it to diagnose leukaemia. Unlike traditional diagnostic methods that may be subjective and dependent on the expertise of medical professionals, deep learning algorithms can provide consistent and objective results. Furthermore, deep learning models have the potential to save time and resources by processing substantial amounts of data quickly and automatically, allowing healthcare professionals to focus on other aspects of patient care.

The use of deep learning for leukemia identification is not without difficulties and restrictions, notwithstanding its possible benefits. Large, annotated datasets are necessary for training deep learning models, and these may not always be accessible for uncommon leukemia subtypes. Because deep learning models are frequently referred to as “black boxes,” interpretability of the models is also a challenge. Very little willingness to be transparent about how they make decisions. Deep learning-based leukemia detection classifiers must carefully take ethical factors like patient privacy and data security into account.

Deep learning has shown promise as a leukemia diagnosis method, with possible advantages including increased consistency, accuracy, and efficiency. Deep learning-based leukemia detection classifiers require further study and improvement to make them more reliable, comprehensible, and morally acceptable before they can be safely and successfully

incorporated into clinical settings. Overcoming challenges related to data availability, interpretability, and ethical considerations will be crucial in realising the full potential using in-depth learning to support healthcare workers in early and accurate leukemia detection, ultimately leading to better patient outcomes.

1.1 Research gap

The diagnosis of leukemia can be a complex task, as it involves classification challenges, reduced accuracy, elevated error rates, and time-intensive procedures, which can all hinder the effectiveness of current systems. However, to address these limitations, a hybrid deep learning algorithm has been introduced, which utilizes advanced techniques to improve prediction accuracy on large datasets. This algorithm can potentially overcome the shortcomings of existing methods and enhance diagnostic capabilities for leukemia cancer detection, leading to more streamlined and dependable processes.

1.2 Major contributions of research

The suggested CNN pre-trained model has made substantial contributions to improving the classification accuracy for leukaemia illness classification. A summary of these contributions is provided below.

- To collect a comprehensive and diverse dataset of blood samples, including both leukemia and non-leukemia samples.
- To prepare the dataset for analysis, remove irrelevant features, handle missing values, and normalize the data to ensure accuracy. To identify relevant features that are associated with leukemia and use them to build a classification model.
- To evaluate different classification algorithms and select the most suitable one.
- The CNN pretrained classifier's main goal is to correctly determine whether or not a patient's blood sample includes leukaemia cells.
- Early detection of leukemia can help improve the chances of successful treatment. Therefore, the classifier should aim to detect leukemia in its initial stages.
- The classifier should be reproducible, meaning that the same results can be obtained from different sets of data.

In this research, the remaining sections are structured as follows:

The analysis of current deep learning-based leukaemia detection algorithms conducted by various researchers is presented in Section 2. The suggested leukaemia detection model's technique is then described in section 3 in more depth. The results of the execution are shown, and Section 4 of the proposal strategy is covered. Section 5 concludes the article by describing its potential scope.

2 Related works

The medical industry is investing in cancer detection, and AI-powered algorithms are showing promise. Studies include a deep learning algorithm for breast cancer detection, an AI-powered algorithm for skin cancer detection, and an algorithm for lung cancer

detection. With further research, these algorithms could improve cancer detection, leading to earlier diagnosis and better treatment outcomes.

A study by researchers at MIT and Harvard Medical School proposed a deep learning algorithm that can detect breast cancer up to 5 years before it can be diagnosed through traditional methods. The algorithm analyzes mammograms and identifies patterns that indicate the presence of cancer. An AI-powered system that can identify skin cancer with an accuracy rate comparable to that of board-certified dermatologists was created by researchers at Stanford University in another study. In their third investigation, a group of researchers from the University of California, Los Angeles (UCLA) suggested an algorithm that had a 94% accuracy rate in identifying lung cancer. The algorithm analyzes CT scans to identify nodules that may indicate the presence of cancer. These studies highlight the potential of AI-powered algorithms to revolutionize cancer detection and improve patient outcomes. With further research and development, these algorithms could become an integral part of the diagnostic process, leading to earlier detection and more effective treatment of cancer.

The studies conducted by Ranjitha P et al. [1] and Neha Agarwal et al. [2] are exemplary demonstrations of the effectiveness of artificial intelligence in the early detection of leukemia. Ranjitha P et al.'s [1] study implemented the K-means technique to distinguish blood cancer, specifically leukemia. The team expertly extracted the necessary features from the data using MATLAB's area props feature. In contrast, Neha Agarwal et al.'s [2] study utilized AI to identify early-stage leukemia through a combination of image enhancement and pattern recognition techniques. The team proficiently employed the TensorFlow library to extract standard deviation and mean values from images, which were then utilized for classification purposes. These studies demonstrate the immense potential of AI-powered algorithms in detecting leukemia at an early stage, which could lead to better treatment outcomes.

Another noteworthy study in the field of leukaemia diagnosis is the analysis of machine learning applications by Haneen T. Salah et al. [3]. They compared different methodologies such as feature extraction techniques, classification algorithms, and assessment criteria to identify the most effective approaches for leukaemia diagnosis. Subrajeet Mohapatra et al. used image processing of microscopic blood pictures to detect acute leukaemia. They employed image analysis, feature extraction, and classification methods, using SVM for classification and fuzzy C-means clustering for image segmentation [4]. Okeb Dese Hakkins Raj created a classification system based on machine learning for identifying leukaemia from blood smear images. They used pre-processing, feature extraction, image segmentation, and classification techniques, including fuzzy C-means clustering for image segmentation [5]. A deep learning-based classifier for the biological diagnosis of leukemia was created by Tawfeeq Shawly et al. [6] utilizing the 8-layer convolutional neural network AlexNet. Gundepudi created a method based on deep learning for detecting acute lymphoblastic leukaemia (ALL) using the ALL-IDB2 dataset, which combined image pre-processing, feature extraction using a deep learning model, and classification using various methods [7].

Sarmad Shafique et al. developed a system for the identification and classification of acute lymphoblastic leukaemia (ALL) and its subtypes using deep convolutional neural networks [8]. Along with XGBoost and AlexNet-based feature extraction, their approach included the use of support vector machines (SVM), decision trees, and k-nearest neighbors (kNN). Nizar Ahmed et al. used convolutional neural networks and other classifiers to distinguish leukaemia subtypes from microscopic pictures. Additionally, they employed data augmentation techniques to increase the size of the dataset and employed 5-fold

cross-validation to compare the accuracy of many classifiers, such as CNN, kNN, decision trees, and SVM [9]. Gautam S. Sailja used a supervised classifier to identify acute myelogenous leukaemia (AML) in blood microscopic pictures, with a methodology that involves applying a median filter to the image before extracting features using K-means clustering and Local Directional Pattern (LDP) algorithms [10].

G Bagasjvara et al. conducted an extensive study of various image processing-based automated detection and classification methods for acute leukaemia based on an analysis of multiple papers, which identified the benefits and drawbacks of various strategies. The authors aimed to provide a thorough analysis of the most cutting-edge methods now available and identify topics for further investigation [11]. Siew Chin Neoh et al. proposed an intelligent decision support system for leukaemia diagnosis [12] by clustering nuclear components using genetic algorithms and stimulation discriminant measures (SDM).

Mu-Chun Su et al. developed a white blood cell classification system that uses machine learning and image processing technologies, utilizing the HSI colour space to categorize photographs and evaluating the effectiveness of three classifiers: hyper rectangular composite neural networks (HRCNNs), support vector machines (SVM), and multilayer perceptron (MLP) [13].

Sonali Mishra et al. suggested a novel technique for diagnosing acute lymphoblastic leukaemia (ALL) using a random forest (RF)-based classification algorithm and a grey level co-occurrence matrix (GLCM). Their method utilized marker-based segmentation (MBS) to separate white blood cells from the background, followed by feature extraction using GLCM and feature reduction using probabilistic principal component analysis (PPCA). The RF method is then used to classify the reduced characteristics [14]. In their article Feature Extraction and Blood Cell Classification for an Automated Differential Blood Count method, Guclu Ongun et al. claimed to have created an automated method for blood cell classification.

Their methodology involved using numerous feature extraction methods, including shape-based, texture-based, and intensity-based methods, and several classifiers, including k-NN, LVQ, MLP, and SVM, for classification [15].

Samabia Tehsin et al. developed a computer-aided diagnosis system for acute lymphoblastic leukaemia (ALL) that uses several feature extractions and pre-processing methods like chain code, bending energy, order statistical filter, and Gaussian filter. The authors also explored several classification methods, including probabilistic neural networks (PNN) and adaptive neuro-fuzzy interactive systems (ANFIS) [16]. In order to diagnose leukemia in blood slides, Luis H.S. Vogado et al. devised a system that combines deep learning and support vector machines (SVM) for classification. They used transfer learning with convolutional neural networks (CNNs) and SVMs and achieved a 99% accuracy rate for the datasets for ALL-IDB1, ALL-IDB2, leukocytes, and CellVision [17]. By optimizing the CNN model's hyperparameters, Maryam Hosseini et al. developed an upgraded CNN for the categorization of leukocytes in microscopic blood images utilizing grid search (GS) and random search (RS). The study made use of the blood cell count and detection (BCCD) dataset [18].

Luis H. S. et al. developed an automated method that extracts features from blood smear pictures using convolutional neural networks (CNNs) to detect leukemia. In addition to principal component analysis (PCA) for feature selection, they used a range of classifiers, such as random forests, support vector machines (SVM), and multilayer perceptrons (MLP), to get high accuracy. The study used the majority voting procedure to get the final conclusion [19]. In their study, Mala Claro et al. looked at a number of CNN models and provided the CNN models for Alert Net, Alert Net-R, and Alert Net-X. Using a range of datasets, including

Leukocytes, ALL-IDB, CellaVision, Atlas, Omid et al. 2014, Omid et al. 2015, ASH-OK, and others, the study evaluates the performance of various models [20].

Anoek Zomaer investigated a possible multi-targeted cancer therapy for glioblastoma (GBM) that included checkpoint blockade with VEGF pathway inhibitors. The methodology included experiments using preclinical mouse models of GBM, tumour imaging techniques, and analysis of gene expression and immune cell infiltration. The authors demonstrated significant improvements in survival rates and immune responses in treated mice [21]. William J. Gradishar's paper "Breast Cancer Research and Therapy," released in 2021, established a forum for the discussion and resolution of continuing disputes in breast cancer research and treatment. The study focuses on the need for a convenient forum for conversation and controversy resolution rather than methodology or analysis [22].

Anoek Zomaer et al. assessed the actual use of trabectedin in patients with soft-tissue sarcoma using information from an Italian national drug-access registry. The authors examined registry data to determine the efficacy and security of trabectedin as a treatment for STS patients [23]. In his work Platelet RNA for early and late cancer detection and localization, Sjors G.J. G explains how blood tests based on tumour-educated platelet (TEP) RNA may be used to identify 18 different forms of cancer. The technique required utilizing blood samples, choosing a biomarker panel, and fine-tuning the algorithm for early- and late-stage cancer diagnosis and localization [24]. Zhiming Gui and others separate the subtypes of the GEO (Gene Expression Omnibus) and TCGA (The Cancer Genome Atlas) datasets using the GSVA (Gene Set Variation Analysis) and CMNF (Consensus Non-negative Matrix Factorization) algorithms. The authors aim to identify NXPH4 (Neurexophilin 4), a novel prognostic and immunotherapeutic marker for muscle-invasive bladder cancer [25].

The article discusses several studies focused on the development of techniques and technologies for the diagnosis and treatment of different types of cancer. Daniel Bottomly et al. [26] conducted a study on acute myeloid leukaemia (AML) to identify biological characteristics of treatment response. G Sherin Ramya et al. [27] developed a technique for cancer cell detection using projective transformation for efficient segmentation, while Lei Cao [28] discussed the development of a convolutional network for detecting abnormal cervical cells during cervical cancer screening. The function of mast cells in non-small cell lung cancer was investigated by Domenico Galetta et al. [29], while the impact of the COVID-19 pandemic on cytopathology lab operations and procedures was assessed by T. V. Ajithkumar et al. [30]. These studies highlight the importance of technology and advanced techniques in the diagnosis and treatment of cancer.

Table 1 compares the several approaches that have been used to detect and categorize leukemia illness using deep learning algorithms. However, the majority of these approaches were found to be inadequate in capturing the complete range of characteristics present in breast cancer imaging. Additionally, due to the high resolution of pathology images, a significant amount of computer resources was required for analysis, which may not be easily accessible. The system was trained on a tiny dataset, and it is noteworthy that this work exclusively focuses on endoscopic ultrasound images.

3 Proposed methodology

Building an extensive dataset of biomedical data, such as images of peripheral blood smears, requires a thorough cleaning process to ensure that it is free of any inconsistencies and is ready for analysis. This process entails applying various pre-processing techniques, such as image scaling, to standardize the data and make it suitable for analysis.

Table 1 Comparative analysis with previous methodologies

Ref. No.	Author's	Methods	Dataset used	Performance Metrics	Major finding	Demerits / Limitations
[31]	Kooi, T., Litjens, G., <i>et.al.</i>	Deep learning with convolutional neural networks.	DDSM and INbreast.	0.82-0.87 AUC.	Outperformed other cutting-edge techniques and reached a high accuracy in diagnosing breast cancer.	Not sufficient to capture the full spectrum of breast cancer imaging features.
[32]	Liu, Y., Gadepalli, K., Norouzi, <i>et.al.</i>	Deep learning with convolutional neural networks.	Camelyon16 Challenge dataset.	0.9776 AUC.	Attained cutting-edge results in the identification of cancer metastases from pathology pictures.	Substantial number of computational resources required for processing high resolution pathology images.
[33, 34]	Xu, Y., Jia, Z., <i>et.al.</i>	Deep convolutional neural network.	MIAS and DDSM.	0.78 AUC.	Achieved high accuracy in detecting breast cancer at an early stage.	Limit its application to other imaging techniques often employed in the diagnosis of breast cancer.
[35]	Li, H., Zhu, Y., <i>et.al.</i>	Convolutional neural networks (CNN).	Clinical dataset of full-field digital mammograms.	0.792 AUC.	demonstrated that CNNs can be used to assess breast cancer risk with high accuracy.	Limited dataset used for training the model.
[36]	Liu, C., Mao, X., <i>et.al</i>	Residual network and attention map.	NCT-CRC-HE-100 K (National Cancer Institute Colorectal Cancer H&E histology images).	0.9395 AUC.	Achieved high accuracy in detecting colon cancer using deep learning with residual network and attention map.	The use of various imaging modalities frequently employed in the diagnosis of colon cancer should be the extent of its applicability.
[37]	Sharma, D., Bhattacharya, S., <i>et.al.</i>	Deep learning with convolutional neural networks.	LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative).	0.86 sensitivity and 0.85 specificity.	Detected lung cancer with excellent accuracy by combining convolutional neural networks with deep learning.	Raises concerns about the generalizability of the proposed approach to larger and more diverse datasets.

Table 1 (continued)

Ref. No.	Author's	Methods	Dataset used	Performance Metrics	Major finding	Demerits / Limitations
[38]	Park, S. H., Kim, M. J., <i>et.al.</i>	Deep learning with digital image analysis.	Liver biopsy images.	0.912 accuracy.	Achieved high accuracy in staging liver fibrosis using deep learning and digital image analysis.	Approach may be limited by the quality of the input images.
[39]	Wu, J., Cao, G., <i>et.al.</i>	Deep learning with convolutional neural networks.	Private dataset of endoscopic ultrasound images.	0.898 F1 score.	Used deep learning and convolutional neural networks to identify pancreatic cancer with excellent accuracy.	The study only focuses on endoscopic ultrasound images.
[40]	Amani Al-Ghraibah and Muhammad Al-Ayyad	The two primary methods that were employed were calculating the average area of WBCs around a bounding box surrounding each red blood cell (RBC) and counting the number of WBCs surrounding RBCs.	combination of "Normal" and "Abnormal" microscopic images.	0.91 and 0.88 accuracy	The Leukemic detection method is based on digital image processing software that is easy to use, quick, and precise.	Limited dataset used. Accuracy is too less
[41]	S Alagu, Ahana Priyanka N, Kavitha G, and Bhoopathy Bagan K	AlexNet, GoogleNet and SqueezeNet, and all features are fused together.	ALL-IDB2 dataset	0.98 accuracy	It offers the most common characteristics for acute lymphoblastic leukemia (ALL) identification.	A 5 m 10s calculating time. Yet, this runtime is a little bit excessive.

Table 1 (continued)

Ref. No.	Author's	Methods	Dataset used	Performance Metrics	Major finding	Demerits / Limitations
[42]	Mohammad Elhasham Billah and Farrukh Javed	Bayesian Convolution Neural Networks (BCNN)	ALL-DB2 datasets	0.98 accuracy	The developed models yield helpful information regarding prediction uncertainty in addition to producing categorizing lymphocyte cells with high accuracy into malignant and non-cancerous categories.	Not dependable for better diagnosis of Acute lymphoblastic leukemia (ALL)
[43]	NASMIN JIWANI et al.	Acute Lymphoblastic Leukemia detection model (ALLDM)	DDS and SDS	0.95 accuracy	Using Computational Deep Learning, a proposal for Pattern Recognition of Acute Lymphoblastic Leukemia	Limited dataset used. Accuracy is too little.
[44]	Batool; Yung-Cheol	A robust model with minimal DL assistance that utilizes EfficientNet-B3	C_NMC_2019 and acute lymphoblastic leukemia (ALL) datasets	0.96 accuracy	EfficientNet-B3 outperformed pre-trained DL classifiers by achieving superior performance and efficiency by utilizing fewer trainable parameters, thereby reducing computational complexity.	There is a need for a workable solution to enhance picture categorization capabilities.
[45]	Pradeep Kumar Das, Biswajeet Sahoo, and Sukadev Meher	The model for detecting acute leukemia based on the Orthogonal SoftMax Layer (OSL)	ALLJDB1, ALLJDB2, and ASH	0.97 accuracy	Additional dropouts and ReLu layers were added to the architecture in order to improve network performance and speed.	Need to improve the performance through features at different resolutions and sampling rates

Table 1 (continued)

Ref. No.	Author's	Methods	Dataset used	Performance Metrics	Major finding	Demerits / Limitations
[46]	Xuekun Song; Ke Liang; Juntao Li	RPCA-Based Noise Reduction	golubEssets exprSets for golub, leukemia data	0.97 accuracy	In terms of gene selection and cancer detection accuracy, the suggested technique outperforms the other six.	Use of dataset was limited. It is not accurate enough. Thus, this is a different approach to obtaining gene groupings that have higher biological significance.
[47]	MALLIGA SUBRAMANIAN et al.	CNN variations with prior training, including MobileNet, VGG-Net, and DenseNet	CT/MRI image dataset	0.92 accuracy	enhanced CNN variations for identifying various cancer kinds.	Limited use of dataset. We must expand the variety of cancer types in our job sets and conduct thorough trials to calibrate multitask learning.

Creating a convolutional neural network (CNN) pre-trained model architecture optimized for the diagnosis of leukaemia necessitates selecting the appropriate architecture, number and type of layers, and activation functions to enhance the model's performance. This step requires significant attention to detail and careful experimentation to ensure that the model is tailored to the specific diagnostic task at hand.

Following the completion of the architecture, training data—which is separated into training and testing/validation sets—is used to train the model. The model continually modifies its parameters during the iterative training phase in order to produce the optimum outcomes.

The model's accuracy is evaluated using the testing/validation data, and adjustments are made to fine-tune the model's settings to optimize its performance.

Overall, meticulous attention to detail and a rigorous and iterative approach to model development and training are essential to creating a highly accurate and effective diagnostic tool for leukaemia and other medical conditions.

It is recommended that the CNN pre-trained model be independently verified or tested on an alternative dataset in order to assess its dependability and generalizability. Interpreting the model's predictions, assessing its effectiveness, and contrasting it with other accepted techniques or benchmarks are all necessary steps in this process. If the deep learning model (Fig. 1) performs well, it can be implemented and incorporated into a healthcare context or other pertinent applications.

3.1 About dataset

To diagnose blood cell-related illnesses, it is often required to identify and describe patient blood samples. There are several medicinal uses for automated techniques that identify and categorize various blood cell subtypes.

This research discusses images from Kaggle, a publicly accessible dataset. (<https://www.kaggle.com/datasets/paultimothymooney/blood-cells?select=dataset2-master>). This dataset comprises 12,500 images of blood cells (JPEG) that have been augmented, along with accompanying CSV files that contain cell type labels. The images are divided into four different folders according to their respective cell types: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. Each subtype contains approximately 3000 images.

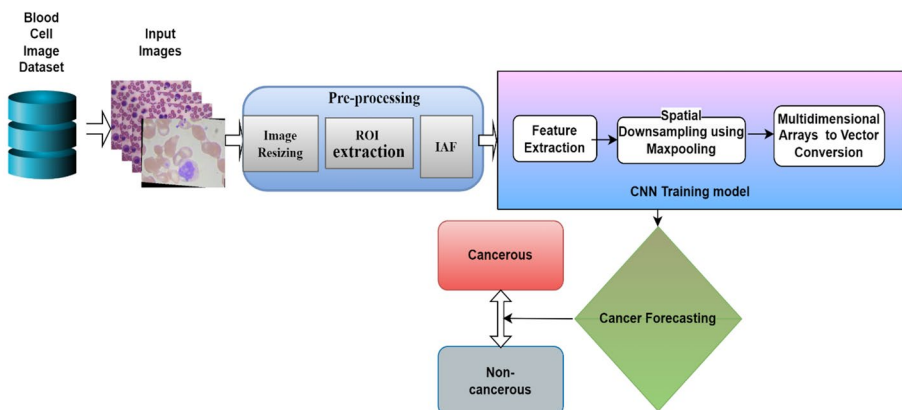


Fig. 1 A proposed schematic CNN pre-trained model for the identification of leukemia malignancy

Furthermore, 410 original pictures (pre-augmentation) with two supplementary subtype labels (WBC vs. WBC) and bounding boxes for each image's cell (JPEG + XML meta-data) are included in the collection. These 410 photos are in the 'dataset-master' folder, whereas 2500 supplemented images, four more subtype labels, and CSV files are in the 'dataset2-master' folder.

For every one of the four classes, there are around 3000 enhanced photographs in the 'dataset2-master' folder, compared to just 88, 33, 21, and 207 images in the 'dataset-master' folder.

Blood-based illnesses sometimes need the identification and characterization of patient blood samples to be diagnosed. There are several medical applications for the automated detection and categorization of blood cell subgroups. The example microscopic cell pictures of cancer patients are displayed in Fig. 2.

The size of the photos is increased by augmenting the blood cell image collection. Twenty percent is utilized for testing and 80 % is used for data training in the suggested model. Table 2 contains information about the training and testing datasets.

3.2 Pre-processing

The cancer detection system's overall efficacy may be compromised by the collected input image's susceptibility to certain sorts of irregularities. To enhance detection accuracy during the classification stage while maintaining original picture quality, the initial step is pre-processing. By first analyzing the input image that was received before proceeding to the next phase, it is possible to remove the inconsistent and noisy pictures that are present in a picture throughout the testing and training phase.

By reducing noisy pixels from each image, image pre-processing is crucial for getting vital information and accurate categorization. The blood cell cancer photos are originally downsized in this stage to 224,224 pixels for models to improve performance. If the original image sizes are used for deep network training, there is a greater chance of visual distortion.

In the pre-processing stage of a blood cell image analysis, images of microscopic cells are first obtained from the blood cell image collection. Anisotropic filtering (IAF) is then used to filter out noise, extract the region of interest (ROI), and resize images from the recorded data to create an enhanced image. During the feature extraction process, the CNN module is used to extract both global and local features. Finally, a pre-trained model predicts whether the blood cell pictures are non-cancerous or malignant based on the extracted features.

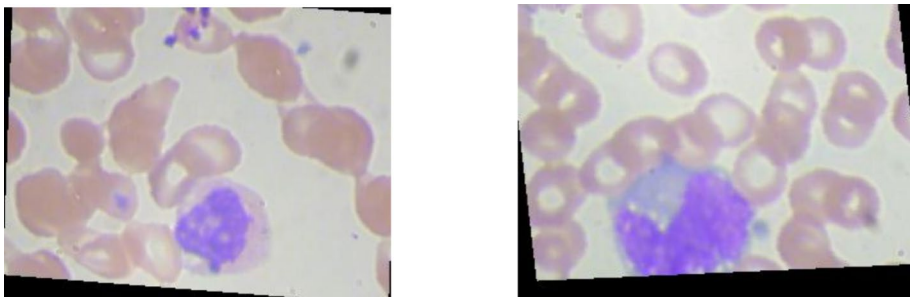


Fig. 2 Patient's microscopic cell images

Table 2 Training and testing data specifications

Dataset used	Number of augmentation images	Number of training samples	The quantity of test samples
Blood Cell Images Dataset	12,500	10,000	2500

One reason to resize a picture is to keep the size uniform across the data set and to use as little memory as possible. It can provide a picture of great quality with subtle changes that are seldom noticeable.

After the image has been downsized, the ROI extraction process is completed throughout the training and testing phases. The image's surrounding text and automated annotations are removed throughout the ROI extraction process. The ROI is defined by a section enclosing the contaminated area in blood cell cancer imaging. The rectangular portion is used to first identify the ROI, after which a mask is made from the rectangle. By using logical indexing, the region outside of ROI was set to zero, and the extracted area was considered.

An image's essential information is preserved by applying filtering techniques. Anisotropic filtering (AF) is a method for eliminating noise that can improve and maintain edge information. The gradient noise can vary over the edge gradient in photographs with a lot of speckle and low contrast, and this technique can be used to identify these fluctuations. These alterations lead to a considerable degradation of edge-related information compared to noise, hence decreasing the effectiveness of the filtering process.

To access the important characteristics of a noisy picture, an IAF technique is used in the suggested methodology. This technique may eliminate speckle noise, which saves important information.

$$A_0 = A.m \quad (1)$$

In eq. (1) where A_0 indicates the speckle-noise-based noise picture m , and the original image is shown as A .

$$m = \frac{A - H}{\sqrt{H}} \quad (2)$$

Equation (2) gives the noise intensity H calculated from picture attributes is indicated above.

$$\sigma = \frac{\sum_{a=1}^M H_a}{M} \quad (3)$$

Where M indicates the total number of iterations and a signifies an image, and where the mean of the noise intensity is indicated as σ in eq. (3). It is possible to compute the kurtosis function as shown in the following eq. (4),

$$k = \frac{\frac{1}{M} \sum_{a=0}^M (H - \sigma)^4}{\left(\frac{1}{M} \sum_{a=0}^M (H - \sigma)^2 \right)^2} - 3 \quad (4)$$

Where k is the kurtosis function. The iteration pausing condition indicates that there must be less association between the picture and the noise class. The method of speckle

suppression keeps the noisy area of the picture closer to the gaussian value (eq. (4)). Equation 5 shows that value k must be 0 in this case.

$$ABS(m - k) \leq 0.001 \quad (5)$$

The presence of low speckles with enhanced edge preservation is indicated when the value k of drops below 0.001. The repetition will end whenever the correlation drops to its lowest value given in eq. (5).

$$\theta A = \frac{\sum_{a=0}^{L-1} \sum_{b=0}^{M-1} a.b.\theta A(a, b) - \sigma A_u \sigma A_v}{\frac{\sum_{a=1}^M (A_{au} - \sigma A_u) (A_{av} - \sigma A_v)}{M}} \quad (6)$$

The correlation of image intensities θA may be assessed using the aforementioned eq. (6).

$$\theta H = \frac{\sum_{a=0}^{L-1} \sum_{b=0}^{M-1} a.b.\theta H(a, b) - \sigma H_u \sigma H_v}{\frac{\sum_{a=1}^M (H_{au} - \sigma H_u) (H_{av} - \sigma H_v)}{M}} \quad (7)$$

The correlation between noise intensities θH may be assessed using the eq. (7) above. The suggested filtering strategy produces the best results when θA and θH exposes the smallest amount of deviation. If k is set to 0, all picture characteristics are eliminated. In order to produce a properly filtered picture, the standard value of k is set to 0.001 in the suggested filtering.

3.3 Model training

Deep learning model training involves optimizing interconnected layers with learnable weights through an iterative process. Training data is fed through the network, generating predictions that are compared to the ground truth to calculate loss. Backpropagation updates the weights to minimize loss. This process is repeated for multiple epochs with hyperparameter adjustments. Regularisation techniques may be applied to prevent overfitting. After training, the model can be used for inference on new data to make predictions based on learned patterns (Fig. 3).

To train the model, these are the sequential steps:

- 3.3.1 Feature extraction
- 3.3.2 Spatial down-sampling using max pooling.
- 3.3.3 Converting multidimensional array to vector

3.3.1 Feature extraction

Extracting features is a pivotal stage in using machine learning algorithms for image classification. The objective of feature extraction is to convert an image's raw pixel data into a collection of noteworthy features that can serve as input to a classifier. Various techniques, including learned features and manually created features, are available for feature extraction in image classification. Experts in the field develop hand-crafted features, which usually

include focusing on a single aspect of the picture, such as a color, texture, or set of edges. The Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Scale-Invariant Feature Transform (SIFT) are a few examples of manually generated features.

The learned features, on the other hand, are those that are automatically extracted from the data using deep learning methods like CNN pre-trained model. This approach uses a hierarchical representation of the input data to teach the model to extract features useful for the picture classification problem. CNN pre-trained model is used to categorize and extract features from the blood samples.

The CNN, which was created to distinguish patterns in two dimensions, is made up of layers of neurons. The CNN pre-trained model uses three distinct types of layers: fully connected, pooling, and convolutional. The network consists of 11 layers overall, excluding the input layer. The input layer receives an RGB image that has been processed separately for each colour channel. Convolution layers refer to the first six levels of the convolution network. In the first two convolution layers, an image is subjected to 16 of 3×3 filters. The second two layers adds 32 of 3×3 filter combinations to the image. And the final two convolution layers apply 64 of 3×3 filters to the picture. The filters are specifically designed to learn and capture various patterns, textures, and structures present in the image. Typically, the filters in the early layers (the first two convolution layers) capture low-level characteristics like edges, corners, and gradients, which act as the fundamental building blocks for more intricate structures.

In the middle layers (the second two convolution layers), the filters capture mid-level features such as shapes, textures, and patterns that are more specific to the input image. Finally, in the last two convolution layers, the filters capture high-level features such as object parts or object representations that are relevant to the task at hand, which in this case is likely to be leukemia cell detection.

The choice of 3×3 filters is a widespread practice in CNN pre-trained models as it is known to be effective in capturing local patterns and features. Each layer's (16, 32, and 64) total number of filters as well as the complexity of the information being captured, determines how well the network can handle it. The ReLU activation function is used by the nonlinear transformation sublayer. The convolution layers of the network shown here employ filters to draw out meaningful information from the RGB-color input image.

Deep learning classifiers frequently utilize the Rectified Linear Unit (ReLU) activation function, particularly ones intended to detect malignancies. The neural network is given non-linearity by the use of a piecewise linear function, which enables it to identify intricate

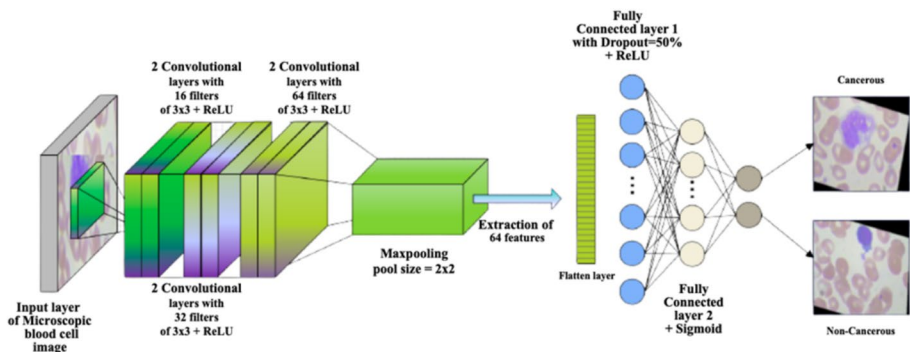


Fig. 3 An architecture diagram illustrating the proposed deep learning model

patterns and produce precise predictions. Equation $f(x) = \max(0, x)$ represents the ReLU activation function, where x is the function's input.

Equation (8) provides the formula for the ReLU activation function:

$$f(x) = \max(0, x) \quad (8)$$

The ReLU activation function takes an input value x and produces an output value $f(x)$. It is active when the input value is positive and inactive when it is zero or negative. The function is represented mathematically by the eq. (9).

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases} \quad (9)$$

where $f(x)$ denotes the function's output and x denotes its input. This function passes through positive numbers unaltered while setting all negative values to 0. Because of its capacity to improve deep neural network training and alleviate the vanishing gradient issue, this straightforward and computationally effective activation function has been extensively employed in deep learning models.

The activation function of the Rectified Linear Unit (ReLU) is utilized by the nonlinear transformation sublayer that follows each convolutional layer. ReLU adds nonlinearity to the network, allowing it to pick up on the intricate and nonlinear interactions between the features in the input picture. This is essential for precise feature extraction and classification in tasks like leukaemia diagnosis. These filters are hierarchically applied to the image, with increasing complexity and abstraction as the layers progress.

Without the need for manual feature engineering, the CNN pre-trained model is built to autonomously extract pertinent features from images. CNNs can be trained to recognize abnormal cells or cellular patterns that are suggestive of leukaemia from medical pictures, such as bone marrow aspirate images or blood smear images. CNNs are particularly good at capturing subtle details that might be challenging for human specialists to notice because they can automatically learn relevant features from the images.

Formally, we first train a sparse autoencoder on tiny $a \times b$ patches x_{small} sampled from some huge $r \times c$ pictures (x_{large}),

$$\text{learning } k \text{ features } f = \sigma(W^{(1)}x_{\text{small}} + b^{(1)}) \quad (10)$$

(Where σ is the sigmoid function), given by the weights $W^{(1)}$ (1) and biases $b^{(1)}$ from the visible units to the hidden units in eq. (10). For every $a \times b$ patch x_s in the large image, we compute.

$$f_s = \sigma(W^{(1)}x_s + b^{(1)}) \quad (11)$$

giving us convolved, a $k \times (r - a + 1) \times (c - b + 1)$ array of convolved features shown in eq. (11).

The best features improved the performance of the classifier and increased the precision, recall, F1 score, accuracy, and detection of medical images, which increased the DL classifier's usefulness.

3.3.2 Down sampling of space using max pooling

In the max pooling sublayer, the image is subjected to a 2×2 filter, which cuts the size of the image in half. One feature for each color channel has now been recovered by the convolution network, and each is represented as a 32×32 array. Using a 2×2 filter, the max pooling

sublayer of the convolutional neural network (CNN pre-trained model) down samples the image by dividing it in half along the width and height dimensions.

The max pooling sublayer is typically applied after the convolutional layer to further down sample the feature maps and reduce their spatial dimensions. In this case, a 2×2 filter is used in the max pooling sublayer, which selects the maximum value within each 2×2 region of the feature maps and discards the rest. This down-sampling operation scales back the feature maps' size by half along both the width and height dimensions, resulting in smaller feature maps with reduced spatial resolution.

$$\text{Output size} = \left(\frac{n + 2p - f}{s} + 1 \right) * \left(\frac{n + 2p - f}{s} + 1 \right) \quad (12)$$

The output will take its final form after that: output size (eq. (12)) is equal to batch shape plus (new_rows, new_cols, and number of filters).

where n is the input's height or breadth, p is padding, s is stride, and f is filter size.

It follows that it is evident that decreasing the output size will result from raising the denominator (the stride).

In this study, the convolution operation of a filter/kernel across an input picture or feature map is determined by the stride function. Typically, a neural network framework's convolutional layer implements the stride function as a parameter. In this experiment, a stride of two is taken into consideration and the kernel size is three. For these parameters, the convolution layer is defined as `conv_layer=nn.Conv2d(kernel_size=3, stride=2, padding=1, in_channels=3, out_channels=16)`. Stride=2 indicates that the filter advances the input picture or feature map by 2 pixels at a time along its height and breadth.

Along with other hyperparameters like filter size and padding, stride selection can have a big impact on the CNN model's properties and performance in image classification tasks. Stride values are usually selected considering the particulars of the work and the desired balance between computing efficiency and spatial resolution.

The purpose of this down sampling operation is twofold: to capture the most important and representative features from the input image, and to reduce the spatial dimensions to enable the network to learn higher-level features and global patterns in subsequent layers. Additionally, down sampling helps in achieving translational invariance, allowing the network to recognize the same patterns in different spatial locations within the image.

3.3.3 Multidimensional Array to vector conversion

Flatten is the eighth layer from the top. By merely concatenating all of the elements in a multidimensional array, the flattened layer reduces it to a single dimension. A 4800 by 4800 one-dimensional grid is the result of this flattening layer. Which results in a transition from the spatial or image-based representation of the data to a vector representation that can be fed into a fully connected layer for further processing. Fully connected layers require inputs in the form of vectors rather than multidimensional arrays, as they treat each input element as a separate feature or input node in the neural network.

3.4 CNN pre-trained model classification

The flattened vector of 4800 input values is applied to the ninth layer, the totally connected ANN with ReLU activation, which uses a dense layer that is fully connected

using the Rectified Linear Unit (ReLU) activation function. The goal of this layer is to provide non-linearity into the network and allow it to recognize intricate patterns in the data by converting the input values into 64 output values using matrix multiplication and element-wise activation using ReLU.

The eleventh layer down is called Dropout. Dropout regularisation is implemented by this layer by repeatedly changing 50% of the input values to zero during training. When a model becomes overly tailored to the training set and is ill-equipped to generalize to new data, this is known as overfitting. Dropout stops this from happening. By randomly deleting input values during training, dropout strengthens the network and prevents overfitting, resulting in a more reliable and generalized model.

Completely linked ANN with sigmoid activation makes up the eleventh and final layer. This layer applies a second completely linked sigmoid activation and employs the 64 input values from the layer before in the function. This layer transforms the 64 input values into two class labels to show the likelihood that the input data belongs to each class in a binary classification problem. The output values between 0 and 1—which are regarded as class probabilities—are squashed by the sigmoid activation function. The final projected class for the input data is then determined by assigning the class label with the highest probability.

In classifiers, especially those used to diagnose leukaemia, A common activation function is the sigmoid activation function. The input is mapped to a probability value between 0 and 1, using a smooth, S-shaped function making it well-suited for binary classification tasks. Mathematical definition of the sigmoid activation function (eq. 13):

$$f(x) = \frac{1}{(1 + e^{-x})} \quad (13)$$

Where $f(x)$ is the function's output and x are the function's input. Values near to 1 indicate a high likelihood of positive class membership, whereas values close to 0 indicate a low chance. The sigmoid activation function calculates the likelihood that the input x belongs to the positive class.

After the features have been extracted, they are fed into a classification algorithm, Neural Networks. The classification algorithm then learns to anticipate the class labels of the input images found on the extracted features. Concisely, feature extraction is a key stage in the use of machine learning algorithms for image categorization. It entails transforming an image's raw pixel data into a set of beneficial traits that can be input into a classifier. The type of data being utilized, and the requirements of the classification task will determine the feature extraction technique to be employed.

First, a convolution network is trained using the input from the training set to identify the weights that create the least amount of error in the two fully connected layers and the weights for the appropriate filters in the three convolutional sublayers.

After that, the convolution network's cross-entropy loss and validation error are computed using the validation set's data. Utilizing the test set's data, one may assess the convolutional neural networks (CNN Pre-trained Model) performance.

3.5 Proposed model analysis using additional methods

A classifier algorithm is a machine learning algorithm that takes input data and categorizes it into one or more classes based on its characteristics. The primary objective of a classifier is to identify patterns in the data that enable it to correctly forecast the class of new, unexpected data.

Several classifier methods may be used, and in order to train one, a dataset containing labeled instances is required. Cancer in blood samples is automatically detected using neural networks. In order to automatically detect cancer in blood samples, neural networks are utilized. Due to their well-known efficiency as a classifier in many real-world applications, neural networks are utilized as a classification tool. The classification algorithms are as follows:

3.5.1 Neural network (proposed)

A machine learning technique called a neural network imitates the composition and operations of the human brain. Networks of nodes or neurons arranged in layers comprise its composition. The layer of neurons underneath the neurons transfers its results to the neurons after the neurons have processed the incoming signals. An input layer, one or more hidden layers, and an output layer make up a network's architecture.

A sequence of mathematical operations is performed in the hidden layers to modify the data that is received in the input layer. The final output is subsequently produced by the output layer using the altered data. In order to enhance the network's capacity for precise prediction, training involves adjusting the weights, or the degree of connectivity between the neurons. To summarise, a neural network is a machine learning system designed to mimic the organization of the human brain. There are layers of interconnected neurons that process incoming impulses and generate output signals.

Regression and classification problems are two applications for neural networks. A neural network's typical architecture consists of an input layer, one or more hidden layers, and an output layer. Each tier that comes after the input layer uses a set of weights and biases that are adjusted throughout training. The final forecast is provided by the output layer, while detailed representations of the input data are found by the hidden layers. Throughout the training process, the neural network adjusts its weights and biases to minimize the difference between the expected and actual output. There may be differences in the architectures of different types of neural networks, including convolutional neural networks (CNNs), feedforward neural networks (FNNs), and recurrent neural networks (RNNs). The simplest type of neural network is called a feedforward network since it is made up of forward-linked layers.

A Convolutional Neural Network (CNN) must have its architecture established. This involves setting up its several layers and parameters. Below is an overview of a CNN:

Let's denote the output of the last fully connected layer as h .

Output Layer:

A single neuron with a sigmoid activation function makes up the output layer. (eq. 14):

$$\hat{y} = \sigma(W \cdot h + b) \quad (14)$$

Where:

$\sigma(x) = 1 + e^{-x}$ is the sigmoid activation function.

W is the weight vector connecting the fully connected layer to the output neuron.

b is the bias term associated with the output neuron.

$\hat{y}$ represents the predicted probability of the input image belonging to the positive class (for containing leukemia cells).

Convolutional Neural Networks (CNNs) are frequently employed for processing images and videos and are specifically developed to learn spatial characteristics from the input data. To extract features from the input data, they employ a convolutional layer, which filters the input data to identify noteworthy features and a pooling layer to reduce the feature dimensionality.

3.6 Cancer forecasting

Following feature extraction, CNN pre-trained model can categorize the medical images as normal or abnormal, which can help in the leukaemia diagnosis process. A sizable dataset of medical images that have been classified as normal or leukaemia-positive can be used to teach CNNs. The CNN pre-trained model can correctly classify new, unseen images as normal or abnormal after learning to recognize patterns in the images that are indicative of leukaemia.

3.6.1 Speed and efficiency

CNNs are suitable for real-time or nearly real-time leukaemia detection because they can rapidly process substantial amounts of image data. In clinical environments, where prompt and accurate diagnosis is essential for patient management, this can be especially helpful. By using the power of GPUs (Graphics Processing Units), CNNs can utilize images in parallel, processing huge image datasets quickly and effectively.

3.6.2 Potential for automation

The detection of leukaemia could be automated with CNN pre-trained model, eliminating the need for physical labour and human error. CNNs can be used in clinical settings to automatically utilize medical images and deliver reliable, unbiased findings after being trained and validated on a sizable dataset. This can aid in lowering inter-observer inconsistency and boosting leukaemia detection's precision and effectiveness.

In conclusion, CNN pre-trained model can accurately classify medical images as normal or abnormal, achieve high accuracy rates, process images quickly, and may even automate the leukaemia detection process, making them useful in the detection of leukaemia cancer. CNN pre-trained model is a promising tool for helping medical workers identify leukaemia early and precisely, which will improve patient outcomes.

3.7 Model testing

The performance of the convolution network is then assessed by computing the cross-entropy loss and validation error using the validation set. Consistently, this assessment procedure is carried out throughout the course of ten epochs. The performance of the convolution network is assessed using the test set once the training for 100 epochs is finished. This iterative approach allows for continuous monitoring of the network's performance and facilitates its learning and improvement over time. This iterative process aids in refining the convolution network and optimizing its performance for improved outcomes in real-world applications.

4 Results and analysis

The project aims to accurately detect cancer by utilizing iterations, loss and accuracy, and a confusion matrix. Conducting a series of twenty iterations, it was observed that the loss decreased with each iteration shown in Fig. 4(a), indicating an improvement in the accuracy of the algorithm. Loss is a measure of the algorithm's accuracy. The model was

successful in lowering the loss value from the first iteration onward, and the value became closer to the minimum with each iteration that followed.

The CNN pre-trained model performed flawlessly throughout the process, resulting in a significant improvement in accuracy.

As shown in Fig. 4(b), Twenty iterations were conducted, and it was found that the loss decreased with each one, demonstrating an increase in the algorithm's accuracy. The accuracy of the algorithm is evaluated by loss. From the very beginning of the process, by lowering the loss value, the model was successful. and with each iteration, it got closer to the minimum loss value. The model ran without error throughout the procedure. The learning rate determines how quickly the algorithm learns from the data and how much it adjusts its parameters in response to errors in the training set. In contrast, if the learning rate is set too low, the process may converge slowly or become stuck in local minima. If the learning rate is set too high, the algorithm may overshoot the ideal solution and fail to converge. In scientific notation, the value $4.999999858590343e-11$ is a very tiny decimal number. It has a step size of $0.00000000004999999858590343$, which is exceedingly small. By reducing the learning rate to this value, the algorithm will take smaller steps during training, which can help it converge to a better solution and avoid overshooting.

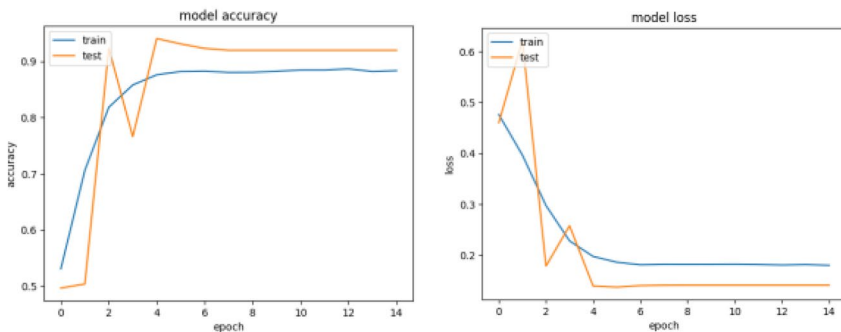
In Fig. 5, the number 0 represent the groups of individuals who do not have cancer, and 1 represents the groups of individuals who have cancer. Employing training, testing, and validation data to examine the accuracy and loss of the suggested classifier model. In Figs. 6(a) and (b), the accuracy and loss performance obtained throughout the training and validation phases are displayed.

```
Epoch 00008: ReduceLROnPlateau reducing learning rate to 4.999999858590343e-11.
Epoch 00008: val_acc did not improve from 0.78205
Epoch 9/10
5216/5216 [=====] - 7s 1ms/step - loss: 0.2243 - acc: 0.9085 - val_loss: 0.4974 - val_acc: 0.7796

Epoch 00009: ReduceLROnPlateau reducing learning rate to 4.999999719812465e-12.
Epoch 00009: val_acc did not improve from 0.78205
Epoch 10/10
5216/5216 [=====] - 7s 1ms/step - loss: 0.2242 - acc: 0.9088 - val_loss: 0.4974 - val_acc: 0.7796

Epoch 00010: ReduceLROnPlateau reducing learning rate to 4.999999546340118e-13.
Epoch 00010: val_acc did not improve from 0.78205
```

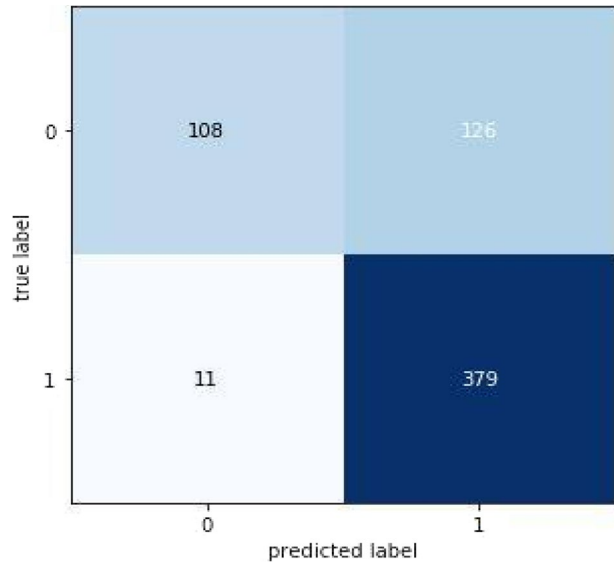
(a)



(b)

Fig. 4 Training and testing loss (a) Iterative algorithmic accuracy (b) Model accuracy and model loss curves

Fig. 5 The trained model's confusion matrix



The accuracy and loss performance of the suggested model may be impacted by modifications to the observed epoch size within the range of 0–100. The training accuracy evaluation and the validation closely resemble the picture that follows. Accuracy increases with increased effort. The length of an era. The recommended CNN pre-trained Model produces accuracy in the range of 95–100% when the epoch size is 100. That the suggested CNN pre-trained model attains the highest accuracy is clearly visible in the image. After the suggested network has been trained using 100 model iterations, the training and validation losses are analyzed. In the 0.03–0.11 range, the model produces losses when the epoch size is extended to 100; these losses become less probable as the epoch size is increased. The suggested CNN pre-trained model has a low error value because the proposed classifier model successfully learned the data. The categorization results for the two samples are shown in Fig. 7.

The suggested CNN pre-trained model's architecture layers are depicted in Table 3, which also provides a depiction of the model's form and parameters as well as its organizational structure.

In this context, recall is considered to be a significant metric. It is crucial to minimize false negatives, as they are included in the denominator of recall and have a significant impact on its value. This is because misdiagnosing a patient with cancer as not having

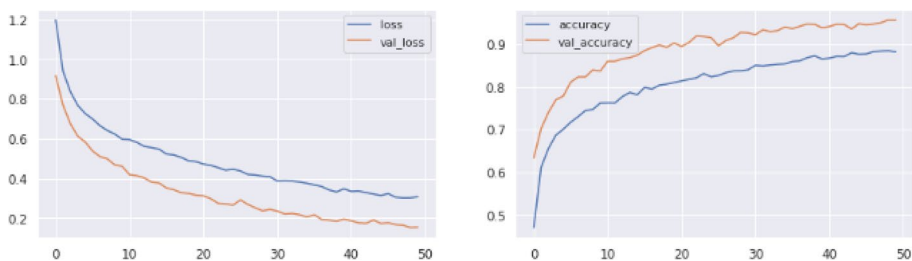


Fig. 6 Performance indicators are the (a) loss and (b) accuracy curves

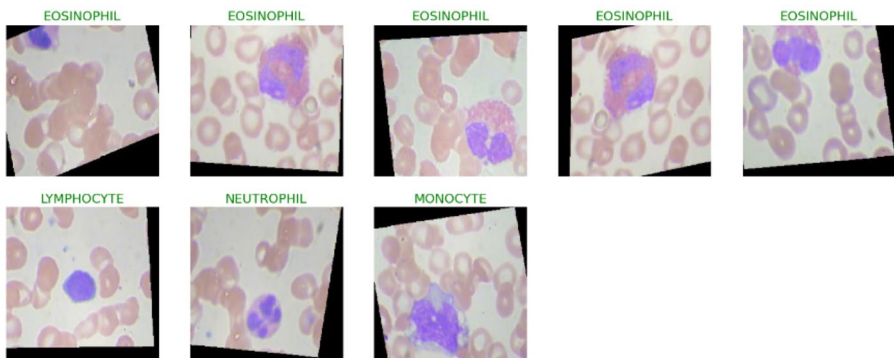
cancer can have severe consequences, making it a primary concern in this research. The objective of this model is to reduce the occurrence of such mistakes made by doctors unintentionally, emphasizing the importance of optimizing recall as a key performance metric. Figure 7(a) shows that the cell types are classified into various classes such as Eosinophil, Lymphocyte, Monocyte, and Neutrophil. Figure 7(b) shows the results of the microscopic cell image as to whether the patient is cancerous or non-cancerous through the cancer detection model.

4.1 Parameters associated with performance

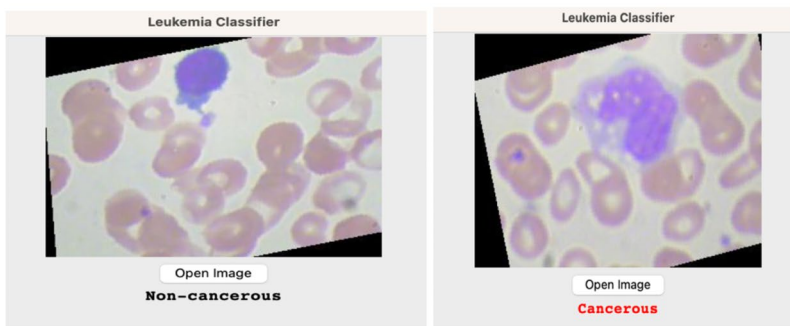
4.1.1 CA (classification accuracy)

The ratio of accurate forecasts to all other predictions may be used to determine the accuracy of a classification model (eq. 15).

$$\text{Accuracy} = \frac{\text{Number of Correct predictions}}{\text{Total no.of predictions}} \quad (15)$$



(a)



(b)

Fig. 7 Images of leukemia cancer (a) Images that have been classified (b) Outcomes of the leukemia cancer detection classifier

4.1.2 F1 (F1 score)

A metric that represents the harmonic mean of accuracy and recall and indicates the trade-off between the two in a classification model (eq. 16).

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (16)$$

4.1.3 Precision

Equation 17 represents the percentage of true positive predictions to all positive predictions, which serves as an indicator of the accuracy with which positive predictions are made in a classification model.

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad (17)$$

4.1.4 Recall (sensitivity or true positive rate)

The ratio of true positive predictions to the total number of observed positive cases (eq. 18) indicates how well a classification model is in identifying positive instances.

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \quad (18)$$

4.1.5 AUC (area under the curve)

The total effectiveness of a classification model is calculated using Eq. 19, which makes use of the area under the curve of the Receiver Operating Characteristic (ROC) curve.

$$AUC = \int ((TP\ Rate) \text{ vs } 1 - (FP\ Rate)) d(Threshold) \quad (19)$$

According to the provided values, the precision of the proposed CNN pre-trained method is 0.989, which is higher than any of the other methods listed. Specifically, the next highest precision value is for k-Nearest Neighbors (kNN) at 0.954, which is 0.035 lower than the proposed method. Tree-based methods such as Decision Trees and Random Forests have a precision of 0.793 and 0.828, respectively, which are significantly lower than the proposed method. Support Vector Machines (SVM) have a precision of 0.102, which is much lower than the proposed method. Finally, Logistic Regression has a precision of 0.013, which is significantly lower than all the other methods. The suggested neural network technique has a better degree of precision than any of the other listed methods, it can be inferred from this.

4.2 ORANGE data mining tool analysis of all algorithms

An open-source data processing toolkit i.e., Orange Data Mining tool is utilized to analyse the various machine learning algorithms to classify the input blood sample image (shown

Table 3 A pre-trained CNN model's architectural layer design

Layer (Type)	Output Shape	Param #
Conv2d (Conv2D)	(None, 150, 150, 16)	448
Conv2d_1 (Conv2D)	(None, 150, 150, 16)	2320
Conv2d_2 (Conv2D)	(None, 150, 150, 32)	4640
Conv2d_3 (Conv2D)	(None, 150, 150, 32)	9248
Conv2d_4 (Conv2D)	(None, 150, 150, 64)	18,496
Conv2d_5 (Conv2D)	(None, 150, 150, 64)	36,928
Max_pooling2d (MaxPooling2D)	(None, 75, 75, 64)	0
Flatten (Flatten)	(None, 36,000)	0
Dense (Dense)	(None, 64)	23,040,064
Dropout (Dropout)	(None, 64)	0
Dense_1	(None, 3)	195

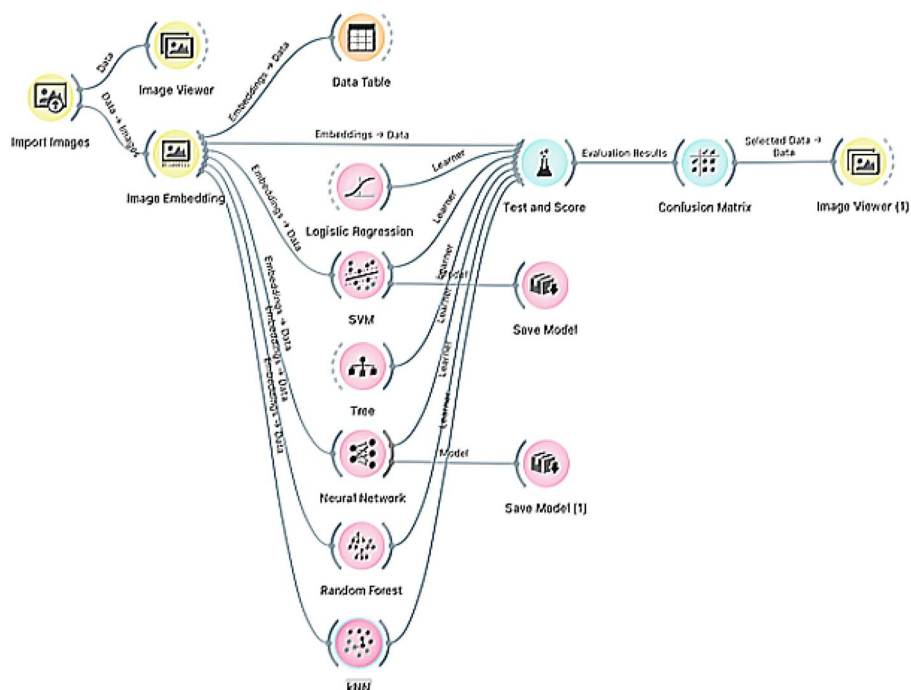
Total params: 23, 112, 339

Trainable params: 23, 112, 339

Non-trainable params: 0

in Fig. 8). The programme functions as a canvas onto which the user may drag widgets for particular data processes before loading their own data. Additional widgets may be used to graphically construct and execute a decision tree for data processing and analysis.

The performance, interpretability, and applicability of diverse methods for resolving real-world issues are all insightfully shown by Orange's examination of a variety of

**Fig. 8** Analysis of several ML algorithms using the orange data mining tool

machine learning algorithms. This application is beneficial for both inexperienced users and seasoned data scientists due to its easy-to-use interface and extensive features.

The graph in Fig. 10 illustrates the performance of various cancer detection methods, including the current methods and the suggested model, for leukemia. The suggested model outperforms the current methods, which include kNN, Tree, SVM, Random Forest, and Logistic Regression. This demonstrates the exceptional effectiveness of the suggested model and shows how it can improve detection system performance. The proposed model's ROC curve analysis utilizing the four-class blood cell dataset is displayed in Fig. 9. The ROC curve is studied in terms of true positive rate (TPR) and false positive rate (FPR). The ideal cut-off has the highest TPR, or sensitivity, together with the lowest FPR, or specificity. ROC curve analysis illustrates the trade-off between TPR and FPR for each possible probability. This demonstrates the capacity to distinguish between two distinct groups and illustrates the efficacy of the leukemia cancer classification system. When the ROC rate is higher, the classification model performs better. The study's findings demonstrate that the suggested model performs better than the alternative models. Using training, testing, and validation data, the accuracy and loss of the proposed classifier model are examined.

For instance, pre-processing, ROI extraction, and scaling are applied to two sample photos. To enhance their quality, the original photos are downsized; ROI extraction is performed to eliminate extraneous symbols or text; and descriptions are given for the pre-processed images. Two distinct classes—cancerous and non-cancerous—are created from the photos. The suggested method yielded a run time of 0.825 seconds.

Table 4 depicts performance analysis of various algorithms to determine their effectiveness in leukemia detection using the ORANGE data mining tool.

Results showcase the potential of cutting-edge algorithms in improving the efficiency of neural networks. Graphical visualization of Table 4 is shown below in Fig. 10.

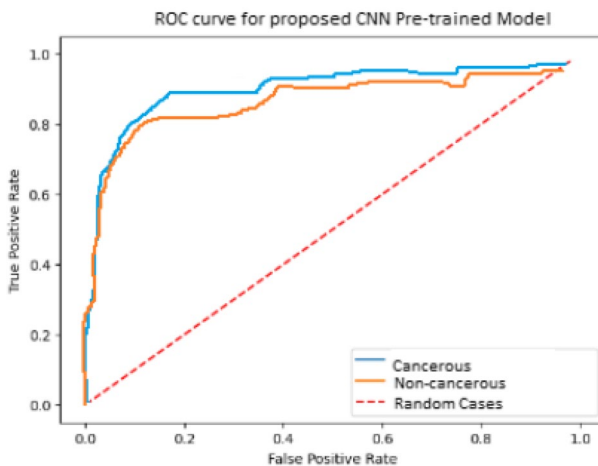
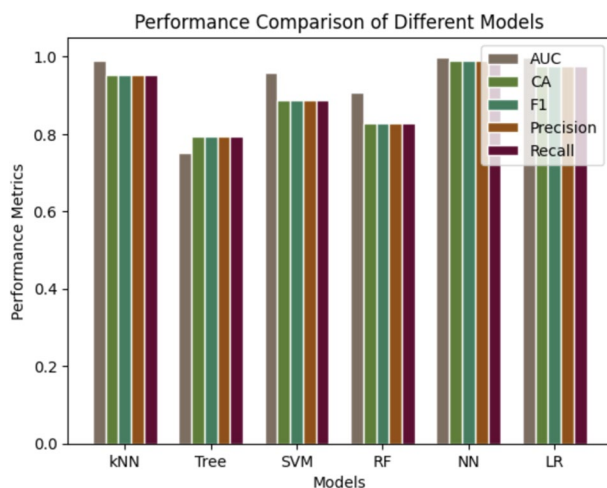


Fig. 9 ROC curve analysis

Table 4 An evaluation of leukemia detection's performance

Model	AUC	CA	F1	Precision	Recall
kNN	0.990	0.953	0.953	0.954	0.953
Tree	0.752	0.793	0.793	0.793	0.793
SVM	0.958	0.887	0.887	0.887	0.887
Random Forest	0.906	0.827	0.827	0.828	0.827
Neural Network (Proposed CNN pre-trained model)	0.999	0.985	0.985	0.985	0.985
Logistic Regression	0.998	0.976	0.976	0.976	0.976

Fig. 10 Evaluation of performance using additional ML methods

5 Performance evaluation

This section includes illustrations of the experimental findings as well as a performance analysis of the proposed CNN pre-trained model. The results of using the proposed model with the Python simulation tool are analyzed. Table 5 displays the experimental configuration for the proposed cancer classification model.

The experiments are paired with the current models to assess the classification model's performance in order to evaluate the presentation of the suggested CNN pre-trained model. The specifics of the suggested model's system setup are shown in Table 6.

The suggested CNN pre-trained model is evaluated against other deep learning techniques already in use on the same dataset, as well as datasets from different sources.

6 Discussion

The potential clinical implications of the deep learning-based classifier (CNN pre-trained model) for leukemia diagnosis are significant. The benefits of the method include its capacity to standardize and automate the diagnosis process, reduce subjectivity, and potentially increase diagnostic precision. By utilizing advanced deep learning techniques, the classifier

Table 5 Configuration for evaluating the suggested CNN pre-trained model

Sl. No	Hyper-parameters	Values
1.	Volume of batch	32
2.	The starting rate of learning	0.001
3.	Algorithm learning	Adam
4.	Epoch (Max. size)	100
5.	The function of activation	CNN
6.	Training data (%)	80
7.	Testing data (%)	20
8.	Validation data (%)	15
9.	Loss-function	Cross entropy

Table 6 System configuration information

Sl. No	Specifications	Setup
1	Pen and touch	Pen and touch input are not available for the display.
2	System type	64-bit OS
3	Installed RAM	4.00 GB
4	Processor	Intel(R) Core(TM) i3-3245 CPU @ 3.40GHz

can provide a standardized and automated approach to leukemia diagnosis, eliminating variability associated with subjective interpretations and human error. This may result in more dependable and consistent outcomes., ultimately improving patient care and outcomes.

Nevertheless, there are issues and restrictions with the strategy that must be resolved. The need for big and varied datasets for deep learning training is one of its drawbacks. How well the model works and how broadly it may be impacted by the occasionally restricted availability of such datasets. Additionally, potential biases in the data used for training the model, such as demographic or geographic biases, could affect the accuracy and fairness of the classifier’s predictions. Ethical considerations, such as patient privacy and data protection, also need to be carefully addressed when implementing deep learning-based classifiers in clinical practice.

In summary, while the CNN pre-trained classifier offers significant benefits in standardizing and automating leukemia diagnosis, some limitations and challenges need to be considered and addressed. Further research and validation are necessary to ensure the accuracy, reliability, and ethical implications of such classifiers in clinical settings.

The suggested CNN pre-trained model may achieve greater performance because to the efficient feature usage, superior training capability, and highly convergent. The suggested model’s error rates may be successfully lowered, and time consumption can be decreased, thanks to good training of the input data.

6.1 Ablation study

Table 7 displays the findings of an ablation study that sought to ascertain the effects of every extra element introduced to the suggested CNN pre-trained model.

Table 7 Ablation research outcomes

Pre-processing	Feature extraction	Feature selection	Classification	Accuracy (%)
No	Yes	Yes	Yes	92.22
Yes	No	Yes	Yes	94.27
Yes	Yes	No	Yes	96.86
Yes	Yes	Yes	Yes	98.5 (Blood Cell images dataset)

When every procedure in the suggested CNN pre-trained model is followed, a better performance of 98.5% accuracy is obtained. When there is no pre-processing phase, very little performance is realized. When the feature extraction process is not done, the accuracy is only 92.22%, and when the feature selection process is not done well, accuracy is only 96.86%.

6.2 A comparison with previous works

Here are a few benchmark methodologies employed for leukemia cancer detection using AI with their explanations and reported accuracy:

- a. “An Efficient Deep Learning Approach to Leukemia Detection” by Raza et al. (2019): This research proposes a deep learning method for leukemia identification in small blood smear images using convolutional neural networks (CNNs). This model provides an accuracy of 97.92%. In contrast, the proposed approach had a 99.2% accuracy which is 1.28% greater than the previous model. It’s interesting to note that the previous model was trained on a limited dataset, which could have contributed to its lower accuracy.
- b. “Automated Acute Leukemia Detection Using Convolutional Neural Networks” by Tsantis et al. (2018): Using CNN as a basis, this study presents the automatic identification of acute leukemia is presented. It appears that the suggested method’s accuracy and sensitivity were lower, at 92.4% and 88.2%, respectively, compared to the suggested model. Additionally, it seems that this model may be facing some challenges in classification.
- c. By Jain et al. (2020), “Leukemia Detection Using Hybrid Deep Learning”: In order to diagnose leukemia, this paper suggests a hybrid deep learning strategy that blends CNN and RNN architectures. The accuracy and sensitivity of the suggested method were both 97.2% and 96.1%, respectively. Here the accuracy and sensitivity of this model are lower than the suggested model, possibly due to the limited dataset used. It’s important to note that a small dataset can limit the ability of a model to accurately classify data.
- d. Leukemia detection using a brand-new framework for deep learning based on capsule network” by Sabour et al. (2018): This article proposes a capsule network-based deep learning method for leukemia detection. Achieved sensitivity of 98.8% and accuracy of 98.1% was the suggested method. It appears that the model has achieved slightly better accuracy, although it was developed using a limited dataset. It is important to note that while a limited dataset can lead to better accuracy, it may not always be a reliable indicator of the model’s performance on new data. It is always recommended to use a larger and diverse dataset to develop a more robust and reliable model.

- e. “Automated detection of acute lymphoblastic leukemia using deep learning-based approach,” Islam et al. (2019). This work proposes an automated deep learning-based approach for the identification of acute lymphoblastic leukemia by combining CNN and RNN architectures. It’s shown that this method achieved comparatively less sensitivity and accuracy scores, with a sensitivity of 98.4% and an accuracy of 98.1%. However, it’s important to note that the model was developed using a limited dataset and is currently facing classification problems. It’s always advisable to use a more comprehensive and varied dataset to develop a more robust and reliable model.

With reported accuracies ranging from 92.4% to 98.6%, these investigations show the potential of AI-based methods for the precise identification of leukaemia as a whole.

Medical samples taken from individuals are used in cancer detection. To determine whether or not these materials are cancerous, tests are conducted. These duties require a lot of manual labour and take a long time to complete. Through the automation of these procedures, the cancer detection tool has sped up the discovery of cancer. This method will make it easier to accurately and efficiently determine whether the input image is healthy or cancerous. The current model has solely focused on the detection of leukemia cancer. However, in the future, the plan is to expand the method to encompass the detection of additional types of cancers. The use of the deep learning-based classifier may be expanded as a result of this development, making it possible to identify various cancer kinds early and accurately. By utilizing deep learning’s capabilities, it will be possible to create a reliable and flexible model that will help both patients and medical professionals by aiding in the detection and treatment of various malignancies. To improve the efficacy and usefulness of the technique in identifying and diagnosing various forms of cancer, more research and development in this field would be essential. Additionally, elements like patient databases, information about medical facilities, and consultation assistance can be added to make it a foolproof, industry-ready system.

7 Conclusions & Future Scope

A deep learning-based classifier for leukemia cancer has been developed in recent research using images of tiny cells. The suggested model can accurately classify leukemia cells, which is a promising step forward in the field. As leukemia is a devastating cancer that affects both physical and emotional well-being, Early detection is crucial for survival. Deep learning-based techniques have improved diagnosis through cell images. Our study accurately diagnoses and treat the disease, improving survival rates and positively impacting patients’ lives. Large and diverse datasets are necessary, potential biases exist, and ethical issues must be considered, among other obstacles that still need to be overcome. It’s also important to note that CNN models may struggle to generalize well to unseen data or data from different populations or settings, which can limit their ability to accurately detect leukemia in diverse populations or diverse types of leukemia. More research and validation are necessary to ensure the effectiveness and accuracy of leukemia diagnosis using this approach. This research showed that the convolutional neural networks (CNN pre-trained model) had an outstanding performance in terms of accuracy and precision, achieving 98.5% in both cases. This is a promising result for the use of deep-learning techniques in improving the diagnosis of leukemia.

The suggested model in the study is limited by the use of a small number of blood cell images. However, the model may be further developed in future work by fusing real-time, massively scalable data sources from hospitals with optimization approaches for fine-tuning the model to get better outcomes. In the future, this study offers promising potential for improving the diagnosis of leukemia through advanced deep-learning techniques. Leukemia patients can receive remote diagnosis and monitoring thanks to the model's integration with telemedicine platforms. This can help reach underserved populations and improve access to quality healthcare. Additionally, it can be utilized for real-time monitoring of leukemia patients, providing continuous feedback on disease progression, treatment response, and potential relapse, which can aid in timely interventions and better patient management.

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Data availability The dataset is available of the following link: (<https://www.kaggle.com/datasets/paultimothymooney/blood-cells?select=dataset2-master>).

Compliance with ethical standards

Conflict of interest The author has not identified any conflicts of interest. The authors must state unequivocally that no proprietary algorithms have been employed.

References

1. Ranjitha P, Duth S (2021) Detection of blood cancer-leukemia using k-means algorithm. In: 2021 5th international conference on intelligent computing and control systems (ICICCS). IEEE, pp 838–842
2. Agarwal N, Agrawal P (2021) Early stage detection of leukemia using artificial intelligence. In: Machine learning for healthcare applications. Wiley, pp 215–224. <https://doi.org/10.1002/jcp.30410>
3. Salah HT, Muhsen IN, Salama ME, Owaidah T, Hashmi SK (2019) Machine learning applications in the diagnosis of leukemia: current trends and future directions. *Int J Lab Hematol* 41(6):717–725. <https://doi.org/10.1111/tid.13169>
4. Mohapatra S, Patra D, Satpathi S (2010) Image analysis of blood microscopic images for acute leukemia detection. In: 2010 international conference on industrial electronics, control and robotics. IEEE, pp 215–219. <https://doi.org/10.1109/ICCSP.2010.5477121>
5. Dese K, Raj H, Ayana G, Yemane T, Adissu W, Krishnamoorthy J, Kwa T (2021) Accurate machine-learning-based classification of leukemia from blood smear images. *Clin Lymphoma Myeloma Leuk* 21(11):e903–e914. <https://doi.org/10.1016/j.compbiomed.2021.104608>
6. Shawly T, Alsheikhy AA (2022) Biomedical diagnosis of leukemia using a deep learner classifier. *Comput Intell Neurosci*. <https://doi.org/10.1155/2022/4574478>
7. Sashank S, Gundepudi V, Jain C, Venkateswaran N (2021) Detection of acute lymphoblastic leukemia by utilizing deep learning methods. In: Machine vision and augmented intelligence—theory and applications: select proceedings of MAI 2021. Springer, Singapore, pp 453–467. https://doi.org/10.1007/978-981-16-3248-4_1
8. Shafique S, Tehsin S (2018) Acute lymphoblastic leukemia detection and classification of its subtypes using pretrained deep convolutional neural networks. *Technol Cancer Res Treat* 17:1533033818802789. https://doi.org/10.1007/978-981-10-8017-3_42
9. Ahmed N, Yigit A, Isik Z, Alpkocak A (2019) Identification of leukemia subtypes from microscopic images using convolutional neural network. *Diagnostics* 9(3):104. https://doi.org/10.1007/978-981-13-7419-5_32
10. Goutam D, Sailaja S (2015) Classification of acute myelogenous leukemia in blood microscopic images using supervised classifier. In: 2015 IEEE international conference on engineering and technology (ICETECH). IEEE, pp 1–5. <https://doi.org/10.1109/ISMAC.2015.7488483>
11. Bagasjvara RG, Candradewi I, Hartati S, Harjoko A (2016) Automated detection and classification techniques of acute leukemia using image processing: a review. In: 2016 2nd international conference

- on science and technology-computer (ICST). IEEE, pp 35–43. <https://doi.org/10.1109/ICoBE.2016.7943136>
12. Neoh C, Siew WS, Zhang L, Todryk S, Greystoke B, Lim CP, Hossain MA, Aslam N (2015) An intelligent decision support system for leukaemia diagnosis using microscopic blood images. *Sci Rep* 5(1):14938. <https://doi.org/10.1038/srep14938>
 13. Su M-C, Cheng C-Y, Wang P-C (2014) A neural-network-based approach to white blood cell classification. *Sci World J*. <https://doi.org/10.1155/2014/676045>
 14. Mishra S, Majhi B, Sa PK, Sharma L (2017) Gray level co-occurrence matrix and random forest based acute lymphoblastic leukemia detection." *Biomed Signal Process Control* 33:272–280. <https://doi.org/10.29011/2575-7675.001085>
 15. Ongun G, Halici U, Leblebicioglu K, Atalay V, Beksaç M, Beksaç S (2001) Feature extraction and classification of blood cells for an automated differential blood count system. In *IJCNN'01. International joint conference on neural networks. Proceedings (cat. No. 01CH37222)*, vol 4. IEEE, pp 2461–2466. <https://doi.org/10.1109/IEMBS.2001.1017168>
 16. Shafique S, Tehsin S (2018) Computer-aided diagnosis of acute lymphoblastic leukaemia. *Comput Math Methods Med* 2018. <https://doi.org/10.1155/2018/7690319>
 17. Vogado LHS, Veras RMS, Araujo FHD, Silva RRV, Aires KRT (2018) Leukemia diagnosis in blood slides using transfer learning in CNNs and SVM for classification. *Eng Appl Artif Intell* 72:415–422. <https://doi.org/10.29011/2575-7675.001102>
 18. Hosseini M, Bani-Hani D, Lam SS (2022) Leukocytes image classification using optimized convolutional neural networks. *Expert Syst Appl* 205:117672. <https://doi.org/10.3390/s22010158>
 19. Vogado LHS, Veras RDMS, Andrade AR, de Araujo FHD, Silva RRV, Aires KRT (2017) Diagnosing leukemia in blood smear images using an ensemble of classifiers and pre-trained convolutional neural networks. In: *2017 30th SIBGRAPI conference on graphics, patterns and images (SIBGRAPI)*. IEEE, pp 367–373. <https://doi.org/10.1109/ISBI.2017.7950664>
 20. Claro M, Vogado L, Veras R, Santana A, Tavares J, Santos J, Machado V (2020) Convolution neural network models for acute leukemia diagnosis. In: *2020 international conference on systems, signals and image processing (IWSSIP)*. IEEE, pp 63–68. <https://doi.org/10.1109/ACCESS.2020.2997841>
 21. Hemminki K, Sundquist K, Sundquist J, Försti A, Liska V, Hemminki A, Li X (2023) Population-attributable fractions of personal comorbidities for liver, gallbladder, and bile duct cancers. *Cancers* 15(12):3092. <https://doi.org/10.3390/cancers15123092>
 22. Chryplewicz A, Scotton J, Tichet M, Zomer A, Shchors K, Joyce JA, Homicsko K, Hanahan D (2022) Cancer cell autophagy, reprogrammed macrophages, and remodeled vasculature in glioblastoma triggers tumor immunity. *Cancer Cell* 40(10):1111–1127. <https://doi.org/10.1016/j.semcancer.2021.12.009>
 23. Sadovskaya AV, Petinati NA, Sats NV et al (2023) Biogenesis of mitochondria in multipotent mesenchymal stromal cells in patients with acute leukemia. *Bull Exp Biol Med* 176:283–289. <https://doi.org/10.1007/s10517-024-06009-y>
 24. Vincenzi B, Napolitano A, Comandone A, Sanfilippo R, Celant S, Olimpieri PP, Di Segni S, Russo P, Casali PG (2023) Trabectedin use in soft-tissue sarcoma patients in a real-world setting: Data from an Italian national drug-access registry. *Int J Cancer* 152(4):761–768. <https://doi.org/10.1002/ijco.26712>
 25. GJG Sjors, Arkani M, Post E, Antunes-Ferreira M, D'Ambrosi S, Daan CL Vessies, Vermunt L et al (2022) Detection and localization of early-and late-stage cancers using platelet RNA. *Cancer Cell* 40(9):999–1009. <https://doi.org/10.1016/j.omtn.2021.12.010>
 26. Gui Z, Ying X, Liu C (2022) NXPH4 Used as a new prognostic and immunotherapeutic marker for muscle-invasive bladder cancer. *J Oncol*:2022. <https://doi.org/10.1155/2022/2142537>
 27. Bottomly D, Long N, Schultz AR, Kurtz SE, Tognon CE, Johnson K, Abel M et al (2022) Integrative analysis of drug response and clinical outcome in acute myeloid leukemia. *Cancer Cell* 40(8):850–864. <https://doi.org/10.1016/j.semcancer.2021.12.004>
 28. Rees MA, Morin CE, Behr GG, Davis JC, Lai H, Morani AC, Parisi MT et al (2023) Imaging of pediatric adrenal tumors: A COG diagnostic imaging committee/SPR oncology committee white paper. *Pediatr Blood Cancer* 70:e29973. <https://doi.org/10.1002/jmri.27756>
 29. Mystica A, Joe G, Ramya S, Dolly RJ (2019) Cancer cell detection using projective transformation for effective segmentation. In: *2019 2nd international conference on signal processing and communication (ICSPC)*. IEEE, pp 198–202. <https://doi.org/10.1109/ICACCE.2019.8742714>
 30. Cao L, Yang J, Rong Z, Li L, Xia B, You C, Lou G et al (2021) A novel attention guided convolutional network for the detection of abnormal cervical cells in cervical cancer screening. *Med Image Anal* 73:102197. <https://doi.org/10.1016/j.media.2021.102156>

31. Kooi T, Litjens G, Van Ginneken B, Gubern-Mérida A, Sánchez CI, Mann R, den Heeten A, Karsssemeijer N (2017) Large scale deep learning for computer aided detection of mammographic lesions. *Med Image Anal* 35:303–312. <https://doi.org/10.1016/j.media.2016.07.007>
32. Liu Y, Gadepalli K, Norouzi M, Dahl GE, Kohlberger T, Boyko A, Venugopalan S et al (2017) Detecting cancer metastases on gigapixel pathology images. *arXiv preprint arXiv:1703.02442*
33. Xu Y, Jia Z, Wang LB, Ai Y, Zhang S (2017) Deep convolutional neural network based early detection of breast cancer using mammography. *J Healthcare Eng* 2017(1274858):1–7. <https://doi.org/10.1155/2017/1274858>
34. Li H, Giger ML, Huynh BQ, Antropova NO (2017) Deep learning in breast cancer risk assessment: evaluation of convolutional neural networks on a clinical dataset of full-field digital mammograms. *J Med Imaging (Bellingham)* 4(4):041304. <https://doi.org/10.1117/1.JMI.4.4.041304>
35. Li H, Zhu Y, Burnside ES, Drukker K (2016) Deep learning in breast cancer risk assessment: evaluation of convolutional neural networks on a clinical dataset of full-field digital mammograms. *J Med Imaging* 3(3):034501. <https://doi.org/10.1117/1.JMI.3.3.034501>
36. Liu C, Mao X, Li Y, Li X (2019) Deep learning-based colon cancer detection using residual network and attention map. *J Med Syst* 43(9):279. <https://doi.org/10.1007/s10916-019-1406-4>
37. Sharma D, Bhattacharya S, Chakraborty C (2017) A deep learning approach for lung cancer detection. *Comput Math Methods Med* 2017:8145071. <https://doi.org/10.1155/2017/8145071>
38. Park SH, Kim MJ, Kim HS, Lee HY, Park J, Kim N, Kim K (Oct. 2018) Liver fibrosis staging with a collagen proportionate area measurement using deep learning and digital image analysis. *Clin Mol Hepatol* 24(4):319–328. <https://doi.org/10.3350/cmh.2017.0091>
39. Wu J, Cao G, Wu J, Yang Y (2019) A novel deep learning approach to pancreatic cancer detection based on endoscopic ultrasound images. *IEEE Access* 7:29196–29203. <https://doi.org/10.1109/ACCESS.2019.2901455>
40. Al-Ghraibah A, Al-Ayyad M (2024) Automated detection of leukemia in blood microscopic images using image processing techniques and unique features: cell count and area ratio. *Cogent Eng* 11:1. <https://doi.org/10.1080/23311916.2024.2304484>
41. Alagu S, Ahana Priyanka N, Kavitha G, Bhoopathy Bagan K (2021) Automatic detection of acute lymphoblastic leukemia using UNET Based segmentation and statistical analysis of fused deep features. *Appl Artif Intell* 35(15):1952–1969. <https://doi.org/10.1080/08839514.2021.1995974>
42. Billah ME, Javed F (2022) Bayesian convolutional neural network-based models for diagnosis of blood Cancer. *Appl Artif Intell* 36:1. <https://doi.org/10.1080/08839514.2021.2011688>
43. Jiwani N, Gupta K, Pau G, Alibakhshikenari M (2023) Pattern recognition of acute lymphoblastic leukemia (ALL) using computational deep learning. *IEEE Access* 11:29541–29553
44. Batool A, Byun Y-C (2023) Lightweight EfficientNetB3 model Based on Depthwise separable convolutions for enhancing classification of leukemia white blood cell images. *IEEE Access* 11:37203–37215. <https://doi.org/10.1109/ACCESS.2023.3266511>
45. Das PK, Sahoo B, Meher S (2023) An efficient detection and classification of acute leukemia using transfer learning and Orthogonal Softmax layer-based model. *IEEE/ACM Trans Comput Biol Bioinform* 20(3):1817–1828. <https://doi.org/10.1109/TCBB.2022.3218590>
46. Song X, Liang K, Li J (2023) WGRRL: a weighted group regularized logistic regression for Cancer Diagnosis and gene selection. *IEEE/ACM Trans Comput Biol Bioinform* 20(2):1563–1573. <https://doi.org/10.1109/TCBB.2022.3203167>
47. Subramanian M, Cho J, Sathishkumar VE, Naren OS (2023) Multiple types of Cancer classification using CT/MRI images Based on learning without forgetting powered deep learning models. *IEEE Access* 11:10336–10354. <https://doi.org/10.1109/ACCESS.2023.3240443>

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